

Task 1: Formulating My World as an MDP

Scenario: Student Evening Routine Before a Quiz

In this task, I model a simple daily-life decision-making problem as a Markov Decision Process, also known as an MDP.

The scenario is based on a student's evening routine before a quiz. Every evening, the student has to decide whether to study, sleep, or hang out. The current condition of the student affects the result of the chosen action.

The objective is to choose actions that maximize the long-term reward.

1. Definition of MDP

A Markov Decision Process is defined as:

$$\text{MDP} = (\mathbf{S}, \mathbf{A}, \mathbf{P}, \mathbf{R}, \gamma)$$

where:

S = State space

A = Action space

P = Transition probability function

R = Reward function

γ = Discount factor

2. State Space

The state space contains all possible situations in which the student can be present.

For this problem, I define three states:

$$\mathbf{S} = \{\mathbf{F}, \mathbf{T}, \mathbf{D}\}$$

where:

Symbol	State	Meaning
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F	Focused	The student is fresh and ready to study
T	Tired	The student has low energy
D	Distracted	The student is distracted or in a social mood

3. Action Space

The action space contains all possible actions that the student can take.

For this problem, the available actions are:

A = {Study, Sleep, Hangout}

Action	Meaning
Study	The student spends the evening studying
Sleep	The student rests or sleeps early
Hangout	The student spends time with friends or does social activities

4. Transition Probability Function

The transition probability function tells the probability of moving to a new state after taking an action in the current state.

It is written as:

$P(s' | s, a)$

This means:

Probability of going to next state **s'** , given that the current state is **s** and the action taken is **a** .

Transition Table

From Focused State F

Current State	Action	Next State F	Next State T	Next State D
F	Study	0.7	0.3	0.0
F	Sleep	0.8	0.2	0.0
F	Hangout	0.2	0.0	0.8

From Tired State T

Current State	Action	Next State F	Next State T	Next State D
T	Study	0.2	0.6	0.2
T	Sleep	0.9	0.1	0.0
T	Hangout	0.0	0.3	0.7

From Distracted State D

Current State	Action	Next State F	Next State T	Next State D
D	Study	0.5	0.1	0.4
D	Sleep	0.7	0.2	0.1
D	Hangout	0.2	0.1	0.7

5. Reward Function

The reward function gives the immediate reward obtained after taking an action in a particular state.

It is written as:

$R(s, a)$

The rewards are chosen based on productivity, rest, and social benefit.

Reward Table

State	Study	Sleep p	Hangout
Focused F	10	3	4
Tired T	2	8	3
Distracted D	6	4	5

Explanation of Rewards

When the student is focused, studying gives the highest reward because the student can study effectively.

When the student is tired, sleeping gives the highest reward because rest helps the student recover and become focused again.

When the student is distracted, studying gives a good reward because it helps the student return to productivity.

6. Discount Factor

The discount factor is taken as:

$$\gamma = 0.9$$

This means future rewards are important, but they are slightly less important than immediate rewards.

7. Value Iteration

Value Iteration is used to calculate the value of each state. The value of a state represents how beneficial it is to be in that state in the long run.

The formula for Value Iteration is:

$$V_{i+1}(s) = \max_a [R(s, a) + \gamma \sum P(s' | s, a) V_i(s')]$$

where:

$V_k(s)$ = value of state s at iteration k
 $R(s, a)$ = immediate reward
 γ = discount factor
 $P(s' | s, a)$ = probability of going to next state s'
 $V_k(s')$ = value of the next state
 $\max_a$ = choose the action that gives maximum value

8. Initial Values

Initially, all state values are taken as zero.

$$V_0(F) = 0$$

$$V_0(T) = 0$$

$$V_0(D) = 0$$

State	V_0
F	0
T	0
D	0

9. Value Iteration Step 1

Using the formula:

$$V_1(s) = \max_a [R(s, a) + \gamma \sum P(s' | s, a) V_0(s')]$$

Since all values in V_0 are zero, the future value term becomes zero.

Therefore:

$$V_1(s) = \max_a R(s, a)$$

For State F

Possible rewards are:

Study = 10
Sleep = 3
Hangout = 4

Therefore:

$$V_i(F) = \max(10, 3, 4)$$

$$V_i(F) = 10$$

Best action from state F is **Study**.

For State T

Possible rewards are:

Study = 2
Sleep = 8
Hangout = 3

Therefore:

$$V_i(T) = \max(2, 8, 3)$$

$$V_i(T) = 8$$

Best action from state T is **Sleep**.

For State D

Possible rewards are:

Study = 6
Sleep = 4
Hangout = 5

Therefore:

$$V_i(D) = \max(6, 4, 5)$$

$$V_i(D) = 6$$

Best action from state D is **Study**.

Values After Step 1

State	V_0	V_1	Best Action
F	0	10	Study
T	0	8	Sleep
D	0	6	Study

10. Value Iteration Step 2

Now we use the values obtained in Step 1:

$$V_1(F) = 10$$

$$V_1(T) = 8$$

$$V_1(D) = 6$$

The formula becomes:

$$V_2(s) = \max_a [R(s, a) + 0.9 \sum P(s' | s, a) V_1(s')]$$

Calculation for State F

Action: Study

From state F, taking action Study gives:

F with probability 0.7

T with probability 0.3

D with probability 0.0

Therefore:

$$Q(F, \text{Study}) = 10 + 0.9[(0.7)(10) + (0.3)(8) + (0.0)(6)]$$

$$Q(F, \text{Study}) = 10 + 0.9[7 + 2.4 + 0]$$

$$Q(F, \text{Study}) = 10 + 0.9(9.4)$$

$$Q(F, \text{Study}) = 10 + 8.46$$

$$Q(F, \text{Study}) = 18.46$$

Action: Sleep

From state F, taking action Sleep gives:

F with probability 0.8

T with probability 0.2

D with probability 0.0

Therefore:

$$Q(F, \text{Sleep}) = 3 + 0.9[(0.8)(10) + (0.2)(8) + (0.0)(6)]$$

$$Q(F, \text{Sleep}) = 3 + 0.9[8 + 1.6 + 0]$$

$$Q(F, \text{Sleep}) = 3 + 0.9(9.6)$$

$$Q(F, \text{Sleep}) = 3 + 8.64$$

$$Q(F, \text{Sleep}) = 11.64$$

Action: Hangout

From state F, taking action Hangout gives:

F with probability 0.2

T with probability 0.0

D with probability 0.8

Therefore:

$$Q(F, \text{Hangout}) = 4 + 0.9[(0.2)(10) + (0.0)(8) + (0.8)(6)]$$

$$Q(F, \text{Hangout}) = 4 + 0.9[2 + 0 + 4.8]$$

$$Q(F, \text{Hangout}) = 4 + 0.9(6.8)$$

$$Q(F, \text{Hangout}) = 4 + 6.12$$

$$Q(F, \text{Hangout}) = 10.12$$

Therefore:

$$V_2(F) = \max(18.46, 11.64, 10.12)$$

$$V_2(F) = 18.46$$

Best action from state F is **Study**.

Calculation for State T

Action: Study

From state T, taking action Study gives:

F with probability 0.2

T with probability 0.6

D with probability 0.2

Therefore:

$$Q(T, \text{Study}) = 2 + 0.9[(0.2)(10) + (0.6)(8) + (0.2)(6)]$$

$$Q(T, \text{Study}) = 2 + 0.9[2 + 4.8 + 1.2]$$

$$Q(T, \text{Study}) = 2 + 0.9(8)$$

$$Q(T, \text{Study}) = 2 + 7.2$$

$$Q(T, \text{Study}) = 9.2$$

Action: Sleep

From state T, taking action Sleep gives:

F with probability 0.9

T with probability 0.1

D with probability 0.0

Therefore:

$$Q(T, \text{Sleep}) = 8 + 0.9[(0.9)(10) + (0.1)(8) + (0.0)(6)]$$

$$Q(T, \text{Sleep}) = 8 + 0.9[9 + 0.8 + 0]$$

$$Q(T, \text{Sleep}) = 8 + 0.9(9.8)$$

$$Q(T, \text{Sleep}) = 8 + 8.82$$

$$Q(T, \text{Sleep}) = 16.82$$

Action: Hangout

From state T, taking action Hangout gives:

F with probability 0.0

T with probability 0.3

D with probability 0.7

Therefore:

$$Q(T, \text{Hangout}) = 3 + 0.9[(0.0)(10) + (0.3)(8) + (0.7)(6)]$$

$$Q(T, \text{Hangout}) = 3 + 0.9[0 + 2.4 + 4.2]$$

$$Q(T, \text{Hangout}) = 3 + 0.9(6.6)$$

$$Q(T, \text{Hangout}) = 3 + 5.94$$

$$Q(T, \text{Hangout}) = 8.94$$

Therefore:

$$V_2(T) = \max(9.2, 16.82, 8.94)$$

$$V_2(T) = 16.82$$

Best action from state T is **Sleep**.

Calculation for State D

Action: Study

From state D, taking action Study gives:

F with probability 0.5
T with probability 0.1
D with probability 0.4

Therefore:

$$Q(D, \text{Study}) = 6 + 0.9[(0.5)(10) + (0.1)(8) + (0.4)(6)]$$

$$Q(D, \text{Study}) = 6 + 0.9[5 + 0.8 + 2.4]$$

$$Q(D, \text{Study}) = 6 + 0.9(8.2)$$

$$Q(D, \text{Study}) = 6 + 7.38$$

$$Q(D, \text{Study}) = 13.38$$

Action: Sleep

From state D, taking action Sleep gives:

F with probability 0.7
T with probability 0.2
D with probability 0.1

Therefore:

$$Q(D, \text{Sleep}) = 4 + 0.9[(0.7)(10) + (0.2)(8) + (0.1)(6)]$$

$$Q(D, \text{Sleep}) = 4 + 0.9[7 + 1.6 + 0.6]$$

$$Q(D, \text{Sleep}) = 4 + 0.9(9.2)$$

$$Q(D, \text{Sleep}) = 4 + 8.28$$

$$Q(D, \text{Sleep}) = 12.28$$

Action: Hangout

From state D, taking action Hangout gives:

F with probability 0.2
T with probability 0.1
D with probability 0.7

Therefore:

$$Q(D, \text{Hangout}) = 5 + 0.9[(0.2)(10) + (0.1)(8) + (0.7)(6)]$$

$$Q(D, \text{Hangout}) = 5 + 0.9[2 + 0.8 + 4.2]$$

$$Q(D, \text{Hangout}) = 5 + 0.9(7)$$

$$Q(D, \text{Hangout}) = 5 + 6.3$$

$$Q(D, \text{Hangout}) = 11.3$$

Therefore:

$$V_2(D) = \max(13.38, 12.28, 11.3)$$

$$V_2(D) = 13.38$$

Best action from state D is **Study**.

11. Final Value Iteration Table

State	V_0	V_1	V_2	Best Action After Step 2
Focused F	0	10	18.46	Study
Tired T	0	8	16.82	Sleep
Distracted D	0	6	13.38	Study

12. Final Policy After Two Iterations

The final policy after exactly two value-iteration steps is:

$$\pi(F) = \text{Study}$$

$$\pi(T) = \text{Sleep}$$

$$\pi(D) = \text{Study}$$

Therefore:

Situation	Best Action
If the student is focused	Study
If the student is tired	Sleep
If the student is distracted	Study

13. Analysis

This MDP models a simple student evening routine before a quiz. The student can be in one of three states: focused, tired, or distracted. Based on the current state, the student can choose to study, sleep, or hang out.

After performing value iteration for exactly two steps, the best action from the focused state is to study. This is because studying while focused gives a high immediate reward and also has a good chance of keeping the student productive.

From the tired state, the best action is to sleep. Although studying may seem useful, the reward for studying while tired is low because the student may not be able to concentrate properly. Sleeping gives a high reward and also has a high probability of moving the student to the focused state.

From the distracted state, the best action is to study. This is because studying gives a good immediate reward and also increases the probability of becoming focused in the future.

Thus, the final policy is:

Focused → Study

Tired → Sleep

Distracted → Study

This shows how Value Iteration balances immediate reward with future benefit. The best action is not always the most comfortable action in the present. Instead, the best action is the one that gives a good reward now and also leads to better future states.